

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I _____ (V. Asha sarthiwika / 192311066) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:V. Asha sarthiwika

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Annexure A – Article Review

Selected Research Article

I selected a peer-reviewed Springer research article published in 2020 that is directly related to Statistical Learning, Logistic Regression, SVM, and medical/diabetes classification. The article is especially suitable for CO 4 because it compares a traditional statistical learning method with several machine-learning methods in a real clinical setting.

Task 1 – Article Summary

(a) Article Citation Details

Authors: Anita L. Lynam, John M. Dennis, Katharine R. Owen, Richard A. Oram, Angus G. Jones, Beverley M. Shields, Lauric A. Ferrat, et al.

Title: *Logistic regression has similar performance to optimised machine learning algorithms in a clinical setting: application to the discrimination between type 1 and type 2 diabetes in young adults*

Journal: *Diagnostic and Prognostic Research*

Year: 2020

Volume: 4

Article Number: 6

Publisher: Springer Nature

DOI: 10.1186/s41512-020-00075-2

Domain: Healthcare / Medical Diagnosis

ML Problem: Classification of patients with diabetes into **Type 1 Diabetes (T1D)** or **Type 2 Diabetes (T2D)**.

[Springer article and full-text page](#)

DOI link

(b) Article Summary

The research investigates whether traditional **Logistic Regression (LR)** can perform as well as more advanced machine-learning algorithms for distinguishing Type 1 diabetes from Type 2 diabetes in young adults. The authors identified a research gap because previous studies comparing statistical models and machine-learning methods often used default model parameters and did not always perform external validation. The authors therefore developed and compared seven models: Logistic Regression, Gradient Boosting Machine, Multivariate Adaptive Regression Splines, Neural Network, K-Nearest Neighbours, Random Forest, and Support Vector Machine. The models used seven clinically meaningful predictors: age, BMI, GADA islet-autoantibodies, sex, total cholesterol, HDL cholesterol, and triglycerides. Models were tuned using grid search and evaluated using nested cross-validation and an independent external validation dataset. The study found that Logistic Regression performed similarly to the more complex machine-learning methods. In external validation, Logistic Regression achieved the highest numerical ROC-AUC of 0.95. The authors concluded that when a medical problem has a small number of strong and well-understood predictors, a simple and interpretable statistical model can be as effective as more complex machine-learning approaches.

Task 2 – Methodology Analysis

(a) ML Techniques Used

The researchers compared **seven classification algorithms**:

1. Logistic Regression
2. Gradient Boosting Machine (GBM)
3. Multivariate Adaptive Regression Splines (MARS)
4. Neural Network (NN)
5. K-Nearest Neighbours (KNN)
6. Random Forest (RF)
7. Support Vector Machine (SVM)

The main statistical-learning technique was **Logistic Regression**, while the other algorithms were used for comparison.

Logistic Regression

Logistic Regression predicts the probability that a patient belongs to the Type 1 diabetes class.

The basic form is:

$$P(Y=1)=\frac{1}{1+e^{-z}}$$

where:

$$z=\beta_0+\beta_1X_1+\beta_2X_2+\dots+\beta_nX_n$$

This makes Logistic Regression particularly useful for binary classification.

Dataset

The study used two UK clinical datasets.

Training Dataset – Exeter Cohort

The original Exeter cohort contained **1,378 participants**. After applying eligibility criteria and removing records with insufficient outcome information, the final training dataset contained:

960 participants

Of these, **135 (14%) were classified as Type 1 diabetes**.

External Validation Dataset – YDX

The external validation dataset came from the **Young Diabetes in Oxford (YDX)** study.

Final validation dataset:

504 participants

including **105 (21%) Type 1 diabetes cases**.

Seven Features

Feature	Type
Age at diagnosis	Numerical
BMI	Numerical
GADA islet-autoantibodies	Binary
Sex	Categorical

Feature	Type
Total cholesterol	Numerical
HDL cholesterol	Numerical
Triglycerides	Numerical

Continuous variables were standardised, GADA was converted into positive/negative categories, and observations with missing predictor values were removed.

Model Preparation

The researchers used:

- Feature standardisation
- Categorical conversion
- Missing-value handling
- Grid-search hyperparameter tuning
- 5-fold cross-validation
- Nested cross-validation
- Independent external validation

The models were implemented using **R 3.5.2 and the Caret package**.

(b) Mapping to CO 4

CO 4 Topic Connection with Article

Inductive Learning Models learn patterns from labelled patient examples.

Decision Trees Tree-based methods were included through Random Forest and Gradient Boosting.

Statistical Learning Logistic Regression is the main statistical model.

SVM Support Vector Machine was one of the seven compared algorithms.

Model Evaluation ROC-AUC, calibration and decision curve analysis were used.

Methodological Strength

The strongest aspect of the methodology is the use of **external validation**. The researchers did not evaluate the models only on the data used for development. They tested the models on a separate YDX dataset. This provides stronger evidence about whether the models can generalise to another patient population.

Methodological Limitation

The study used only **seven predictors**. Although these were clinically meaningful and helped reduce overfitting, the authors acknowledged that machine-learning algorithms can potentially benefit from larger numbers of variables and observations.

Task 3 – Results & Critical Evaluation

(a) Key Results

The study found that all seven models performed strongly.

During internal validation:

$$\text{ROC-AUC} \geq 0.94$$

For external validation, performance decreased slightly, but all models still achieved approximately:

$$\text{ROC-AUC} \geq 0.942$$

Logistic Regression achieved the highest numerical external validation ROC-AUC:

$$\text{ROC-AUC} = 0.94$$

The authors also found that Logistic Regression had good calibration and decision-curve performance.

Comparative Table

Model	Internal Validation	External Validation	Observation
Logistic Regression	≥ 0.94	0.95	Excellent
SVM	≥ 0.94	High	Excellent
Neural Network	≥ 0.94	High	Excellent
GBM	≥ 0.94	High	Excellent
Random Forest	≥ 0.94	High	Excellent
KNN	≥ 0.94	High	Excellent
MARS	≥ 0.94	High	Excellent

The article primarily reports **ROC-AUC, calibration and decision-curve analysis**, rather than focusing on accuracy and F1-score.

(b) Critical Evaluation

Are the Results Realistic?

The results are strong, but they are reasonably supported by the experimental design.

The researchers used:

- Hyperparameter tuning
- Nested cross-validation
- External validation
- Calibration analysis
- Decision curve analysis

Therefore, the reported performance is more convincing than a result obtained from only a single train-test split.

However, the external validation performance was slightly lower than internal validation, demonstrating that performance can decrease when a model is applied to a different population.

Potential Dataset Bias

One important limitation is that all participants included in the analysed datasets were **white European participants**. Therefore, the results may not automatically generalise to other ethnic or geographic populations.

Another limitation is that the external dataset differed from the training population. For example, the YDX population was younger and had different BMI and Type 1 diabetes prevalence.

Additional Experiments

The research could be strengthened by:

1. Testing the models on datasets from different countries.
2. Including more diverse ethnic populations.
3. Using a larger number of clinical variables.
4. Performing prospective clinical validation.
5. Comparing additional fairness metrics.
6. Evaluating calibration separately across demographic groups.
7. Testing the models after several years to determine whether performance remains stable.

(c) Real-World Applicability

The technique can be deployed in:

- Hospitals
- Diabetes clinics
- Electronic Health Record systems
- Clinical decision-support systems
- Telemedicine platforms
- Healthcare mobile applications
- Primary-care screening systems

For example:

Patient Information



Age + BMI + GADA + Sex



Logistic Regression



Probability of Type 1 Diabetes



Clinical Decision Support



Doctor's Final Decision

Industry Challenges

1. Data Availability

Healthcare organisations may have incomplete or inconsistent patient records.

2. Data Privacy

Patient information must be securely stored and processed.

3. Generalisation

A model developed using UK patients may not work equally well for patients in India or other populations.

4. Explainability

Healthcare professionals need to understand why a model produced a particular prediction.

5. Clinical Validation

High AUC alone does not prove that the model is safe for clinical use.

6. Model Maintenance

Medical populations and clinical practices change over time, so models need continuous monitoring and validation.

Task 4 – Reflection & Learning Outcome

(a) Three Key Insights

Insight 1 – Statistical Learning Can Compete with Complex ML

The article showed that Logistic Regression can perform as well as more complex algorithms when there are a small number of strong predictors.

CO 4 Connection: Statistical Learning / Logistic Regression.

This taught me that a more complex model is not automatically a better model.

Insight 2 – External Validation Is Very Important

The models performed slightly worse on the external validation dataset than during internal validation.

This demonstrates that a model can perform very well on development data but may lose performance on a different population.

CO 4 Connection: Inductive Learning and Model Evaluation.

It helped me understand the importance of testing whether a learned model generalises to unseen data.

Insight 3 – Accuracy Alone Is Not Enough in Healthcare

The study considered ROC-AUC, calibration and decision curve analysis rather than relying only on accuracy.

CO 4 Connection: Statistical Learning and Evaluation Metrics.

In healthcare, a model must not only classify correctly but should also produce useful and reliable risk estimates for clinical decision-making.

(b) Proposed Extension

Proposed Experiment: Test the Models on a Diverse Multi-Country Diabetes Dataset

I would extend the research by evaluating Logistic Regression, SVM, Random Forest and Neural Network models using diabetes datasets collected from different populations, including an Indian patient population.

The experiment would follow:

Original UK Dataset

+

Indian Dataset

+

Other Population Dataset

↓

Train Models

↓

External Validation

↓

Compare AUC + Calibration

↓

Fairness Analysis

Justification

The original study mainly used white European participants. Testing the models on more diverse populations would determine whether the learned patterns generalise across demographic groups.

The expected impact would be:

- Better understanding of model generalisation
- Detection of population-specific bias
- More reliable clinical deployment
- Improved fairness
- Better evidence for real-world healthcare applications